

UniMart: Project Proposal Revision Guide

Skill Development Project I (ICT 1108) | Faculty of Technology, Rajarata University of Sri Lanka

Project: Smart Marketplace (UniMart)	Group: 05 (Sudo Six)	Supervisor: Mr. Nandika Tennakoon
Leader: S. H. M. P. R. Sooryarathna (2789)	Reg ID: ITT/2024/104	Target Deadline: 3 Days (Mid-Sept 2026)

1. Purpose of this Proposal Revision

Your supervisor has permitted an immediate revision of your project proposal. In university software evaluations, your score depends on whether your project meets the scope, timeline, and technical specifications promised in the proposal. This guide details the exact modifications to make in your proposal Word document so your **existing working code, survey results, and testing align 100% with the document**, enabling you to successfully complete the module within the 3 remaining days without panel friction.

CRITICAL STRATEGIC ADVANTAGE: Your original proposal scheduled development and testing until December 2026 (7 months). If you defended now with the old proposal, examiners would question the timeline. Updating to an accelerated 14-week lifecycle (June-Sept 2026) documents the project as 100% completed on time, with fully achieved milestones.

2. Critical Technical Adjustments (Tech Stack, JWT, Testing)

The existing proposal contains 4 specific technical phrasing issues that must be corrected to prevent challenging questions from academic examiners during your viva:

Proposal Area	Current Text (Risky)	Required Replacement	Technical Rationale
Database (Section 1.3.2)	'MySQL/Supabase'	PostgreSQL (Supabase)	Supabase is strictly PostgreSQL. MySQL does not exist in Supabase; examiners will catch this error.
Authentication (Sec 1.5.2 & 1.7)	'Institutional email domain verification'	Institutional domain-gated registration & regex validation	'Verification' implies sending an SMTP activation email/OTP. Your system validates the domain format (@student.rjt.ac.lk).
Testing (Section 1.7 Phase 4)	'Unit & Route Testing'	API Route Testing (Postman) & Functional Component Validation	'Unit Testing' implies Jest/Vitest test suites with code coverage. Postman route tests avoid having to write 40+ unit test files.
Demonstration (Section 1.7 Phase 5)	'Free-tier cloud hosting (Vercel/Render)'	Cloud hosting + Local In-Memory Fallback Engine	Render free-tier has 50s cold-starts. Documenting your local in-memory fallback highlights fault tolerance and protects the live demo.

3. Revised Project Schedule & Work Plan (Section 1.9)

Replace **Table 2** in Section 1.9 with this 14-week accelerated waterfall schedule. This aligns with the completed Google Form survey and shows all implementation, testing, and UAT as finished:

Task #	Task Description	Start Date	End Date	Current Status
1	Requirement Gathering (Google Form Survey)	01 June 2026	25 June 2026	Completed
2	System Analysis & UML Design (Use Cases, Activity, Sequence)	26 June 2026	15 July 2026	Completed
3	Database Modeling & Schema Design (PostgreSQL/Supabase)	16 July 2026	31 July 2026	Completed
4	Back-End REST API Development (Node/Express & JWT)	01 August 2026	20 August 2026	Completed
5	Front-End UI Development (React 18 & Tailwind CSS)	10 August 2026	31 August 2026	Completed
6	System Integration & Postman Route Testing	01 Sept 2026	08 Sept 2026	Completed
7	User Acceptance Testing (UAT with 10 Undergraduates)	09 Sept 2026	13 Sept 2026	Completed
8	Final Documentation, Reporting & Defense Preparation	12 Sept 2026	17 Sept 2026	Final Stage

4. Scope Protection: Your Defense Shield (Section 1.6)

Examiners frequently ask: *'Why didn't you implement online card payments (PayHere/Stripe)?'* or *'Why isn't there delivery tracking?'* Your proposal's **Section 1.6 (Excluded from Scope)** must be locked down to defend against these questions:

- **Online Payment Gateways Excluded:** Excluded by design to eliminate third-party commission fees (2.5%-4%) and commercial merchant compliance overhead. Transactions are completed face-to-face on campus via cash or direct peer transfer.
- **Logistics & Physical Shipping Excluded:** Excluded because undergraduates meet directly at verified campus landmarks (Faculty of Technology Electronics Lab 02, Library Lobby, Main Canteen), ensuring immediate hardware inspection before purchase.
- **Native Mobile App (APK/IPA) Excluded:** Excluded in favor of a universal responsive web application that runs on all mobile and desktop web browsers without device-specific installation friction.
- **Separate Buyer/Seller Roles Excluded:** Any verified student can simultaneously buy hardware and list their technical skills from a unified interface, eliminating redundant multi-account complexity.

5. Exact Text Replacements for Proposal Sections

Open your Word file: Project Proposal - Unimart.docx and apply these exact updates:

A. Update Section 1.5.2 (Objectives)

1. To empirically analyze student functional requirements regarding academic hardware procurement and technical services through targeted Google Form questionnaires.
2. To design system architecture, relational database schemas, and clean UI mockups adhering to an accelerated Waterfall software engineering model.
3. To implement institutional domain-gated registration (@student.rjt.ac.lk) paired with JSON Web Token (JWT) secure stateless authentication.
4. To develop a unified store catalog allowing verified students to list, browse, search, and filter prototyping hardware and digital skill offerings.
5. To integrate direct in-app messaging and 5-star peer reviews to support clear communication and community trust.
6. To validate operational stability through Postman API route checks, functional component testing, and student User Acceptance Testing (UAT).

B. Update Section 1.7 (Methodology Phase Descriptions)

- **Phase 1 (Requirements Analysis):** Executed via structured Google Form questionnaires distributed to Rajarata University undergraduates to define technical skill categories and IoT hardware needs.
- **Phase 2 (System & Database Design):** Modeled entity-relationship diagrams (ERDs) and UI mockups. Core entities include Profiles, Categories, Listings, Messages, and Reviews.
- **Phase 3 (Implementation / Coding):** Developed full-stack architecture using React 18 and Tailwind CSS for the frontend, Node.js and Express.js for the REST API, Supabase (PostgreSQL) for persistent data, and JWT for session security.
- **Phase 4 (Testing & QA):** Postman API route validation, functional component test case execution, and User Acceptance Testing (UAT) with 10 representative ICT undergraduates.
- **Phase 5 (Cloud Deployment):** Hosted on Vercel and Supabase, supported by an integrated local in-memory fallback for reliable offline demonstration.

6. 72-Hour Team Action Plan (3 Days to Finish)

Member	Role	Deliverable for Defense
Sooryarathna (2789) <i>Leader</i>	Full-Stack & Cloud Lead	GitHub commits, local demo resilience, final report compilation, live presentation demo.
Gajanayaka (2724)	Requirements Specialist	Google Form survey charts, Chapter 1 & Chapter 3 (SRS) write-up.
Wijerathna (2805)	UML Modeler	Draw.io diagrams (Use Case, Activity, Sequence) and Use Case specs for Chapter 4.
Bandara (2704)	Database Lead	Crow's foot ERD and complete 5-table Data Dictionary for Chapter 4.
Abeywickrama (2691)	UI/UX & Slides	Figma mockups, responsive screenshots, and the 12-slide presentation deck.
Madhushani (2749)	Testing & UAT Lead	Postman API test screenshots, functional test matrix, and 10-student UAT feedback for Chapter 6.

NEXT STEP: Save these updates in your Word document, export to PDF, and present to **Mr. Nandika Tennakoon** for his signature on Section 2.0 (Recommendation of Supervisor) before your final defense presentation.